

Project Design Phase
Proposed Solution Template

Date	10 July 2026
Team ID	
Project Name	Smart Lender – AI-Powered Loan Approval Prediction System
Maximum Marks	2 Marks

S.No.	Parameter	Description
1	Problem Statement	Applicants face delays and uncertainty in loan approval due to manual verification and inconsistent criteria; lenders struggle with workload and inaccurate decisions.
2	Idea / Solution Description	A Flask web application that predicts loan approval using a KNN classification model (selected after comparing Decision Tree, Random Forest, KNN and XGBoost), and augments the prediction with eligibility %, confidence score, risk level, reasons and recommendations.
3	Novelty / Uniqueness	Combines a compared, best-of-four ML model with an explainability layer built from application logic, rather than presenting a bare Approved/Rejected label.
4	Social Impact / Customer Satisfaction	Faster, more consistent and more transparent loan evaluation improves trust for applicants and reduces manual workload for loan officers.
5	Scalability of the Solution	The trained model, encoder and scaler are saved as reusable artifacts (model.pkl, encoder.pkl, scaler.pkl), allowing the Flask backend to be redeployed or the model retrained without redesigning the application.